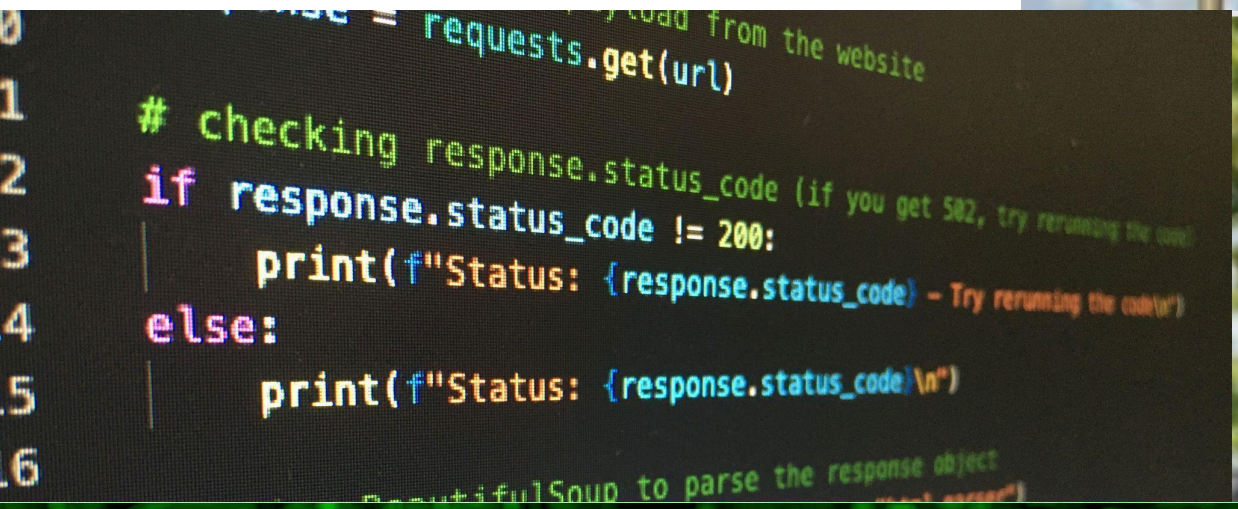
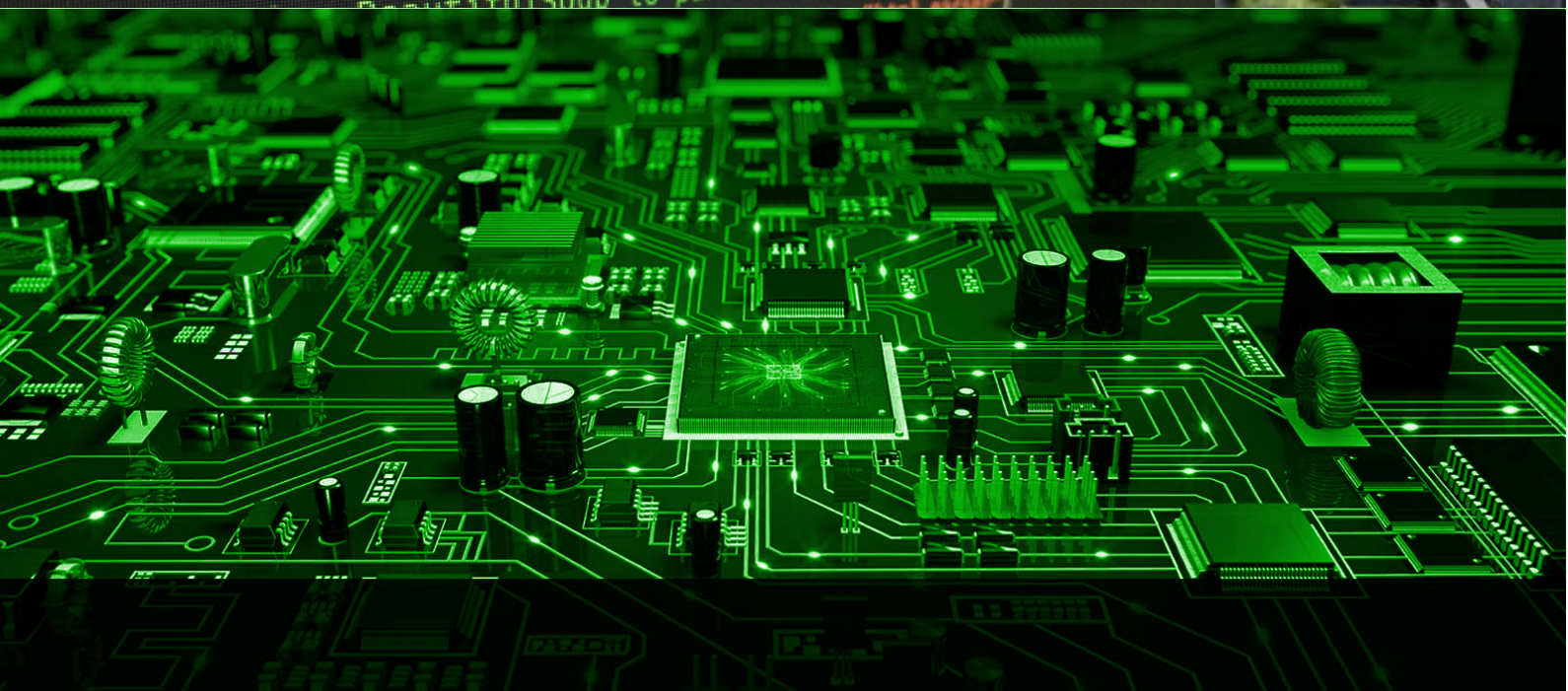
ENG1013: Engineering smart systems

Traffic Control System Project

Project Specifications 2026





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1 Preface

“Smart systems” are critical to the future of engineering, as they address economic, environmental, and societal issues. They are utilised across many industries, including energy supply, healthcare, agriculture, manufacturing, security, entertainment, and logistics. Some examples of smart systems are automatic lighting and climate control in buildings, responsive traffic light controllers, liquid level monitoring in agricultural and industrial settings, intruder alarm systems, driver assistance systems in automotive vehicles, and personal healthcare devices. In smart systems, sensors and timers are often used to detect certain events. Pre-programmed software is then used to perform analysis. This analysis can be used to make decisions and perform actions when combined with historical data.

In this project, you will work with your team to apply your software, electrical, electronic and mechatronics engineering knowledge learned throughout ENG1013 to build a **responsible traffic control system.** This type of traffic control system will control vehicular and pedestrian traffic and is similar to those found at road intersections and pedestrian crossings to help control and manage traffic flow. These systems can be activated based on timing devices, the detection of vehicles, or the press of a button.

In the course of working on the project, it is completely normal for engineering design tasks to feel overwhelming or confusing at the beginning. You're stepping into a process that involves solving complex problems, often without all the answers in front of you. This is an exciting challenge, and here's an important tip to help you succeed: **ask questions—lots of them**.

When you're working on a design project, think of yourself as a detective. Your "client" (whether it's your professor, a real-world client, or even a project brief) has a vision for how the system should work, but they might not give you every detail upfront. Part of your job is to ensure that you’ve understood (and obtained!) all the critical information so that you (and your team) can build an engineering solution that correctly meets the requirements.

Here are a few supportive reminders for approaching this:

- **Confusion is part of the process**: Feeling uncertain is a sign that you're in the middle of learning something new and challenging. Instead of being frustrated, use that feeling to start digging deeper.
- **Questions lead to clarity**: If something doesn't make sense, or if you're not sure about a requirement, ask! There’s no such thing as a “silly question” in engineering design. Often, asking the right question can save you hours of effort and guide your team toward better solutions.
- **It's a collaborative effort**: Engineering design isn’t done in isolation. Asking questions isn’t just for you—it also helps your client clarify what they want you to build, which might not be obvious at all.
- **Learn to listen and observe**: Sometimes, answers you need aren’t just in the words a client says—they’re in how they say it, the constraints they emphasize, or even the problems they didn’t mention.

Remember, every great engineer started where you are right now. It’s okay to feel confused—embrace it as an opportunity to learn. By asking thoughtful questions, you’ll build not only your understanding but also your confidence as you tackle these exciting design challenges. You’ve got this!

2 Project Resources

2.1 Index

The last page in your lab manual contains a list of activities that link to various components of the project. You also can find component videos and datasheets on Moodle, under Learning > Project Resources > Guides and Datasheets.

2.2 Standards

You must follow these “standards” when formatting your code and producing circuit diagrams. Standards are important to ensure that your designs can be understood and reproduced by other engineers.

Circuit Components Library (Draw.io) (Right Click + save to your computer > Go to "Draw.io" > File > Open Library From > Device > navigate to the file you downloaded)

Coding Standards Document

Circuit Standards Document

2.3 Help and Support

There are many avenues available for you and your team to get help with the project.

- You can ask your manager questions during the weekly practical class.
- You can ask your PASS leader questions during the weekly PASS hour that follows your weekly 2h practical class.
- You can ask questions on the Ed forum. If the question is specific to assessable code (eg debugging) please submit it as a private query, as per the forum guidelines.

Remember that no question is a stupid question. Your manager (and all the other managers) want you to succeed and are always happy to guide you.

Students have access to wire and wire strippers in the Week 10 practical, which is a dedicated session for project help and wire stripping.

Please note that support for any milestone will terminate at 6pm on the deadline.

3 Project Restrictions

Your team has been asked to design, build, and demonstrate a simplified traffic control system in this project. Your system will consist of multiple subsystems, and the client has split these subsystems into groups of *required* features, *general* features and *integration* features. There is also an additional *complete integration* point value available. Most subsystems will contain multiple *general* and *integration* features which your team can choose to implement, which you can read about in the milestone deliverables section, and in the system features section below. Testing will **only** consider the case of a single vehicle passing through the system, which may or may not be overheight. Teams should select an appropriate scale for the project.

The project is graded out of 260 points, with the final point value being scaled based on your individual contributions as a team member, and your final viva interview. Your mark is scaled according to the Weighted Scaling Factor to form 26% of your total mark for ENG1013. This project is not part of any hurdle requirement. You can find rubric feedback under the relevant link in Moodle under 'Learning > Project Resources > Results' which shows your team's performance in each milestone against each assessed item.

3.1 Hardware Requirement

You are permitted to use any and all of the hardware components provided in **all** of your team's supplied kits (i.e. all 5 combined), plus single core wire and additional breadboards. However, you may use only **ONE** Arduino to attain maximum points for a fully integrated system. If your team isn't aiming to attain points from the 'integration features', you can use multiple Arduinos - one for each subsystem - note that only one Arduino can be 'active' and running at once. You may not use any other hardware components. You may use more shift registers, op-amps, timers and MOSFETs than provided as long as they are the same component.

3.2 Software Requirements

You may develop your software in any Integrated Development Environment (IDE); however, support will only be provided by demonstrators in VSCode. You may only use the following additional packages within Python 3.10.x:

- Pymata4
- Time
- Math
- Matplotlib
- Random

The following packages are permitted, but demonstrators will not provide support:

- NumPy
- CSV

All other packages, Classes and assert statements are not to be used. If found, they will be commented out for marking.

Students *may* use generative AI but must cite and provide a record of generative AI use (as a pdf) in their submission.

3.3 Academic Integrity Warning

This is a team project. This means that all members of the team have equal responsibility to review work submitted, and ensure that all members comply with the academic standards required of you as a student of Monash University.

You are **permitted to**:

- Use generative AI to assist them in improving paragraphs of the report that is written by you (such as vocab, structure or grammar).
- Use generative AI to assist them in code that is written by you (such as structure or standards), to understand circuit states (or debug), or verify calculations made. Students can use generative AI to understand how code syntax or a circuit works.
- Work with your team-mates
- Seek help from your team-mate
- Seek help from unit staff in-class or via the ed forums
- Use the Internet to look up resources. If you use work that isn't your own, ensure that you provide a citation to the source. If you are unsure if you can use said work, please seek advice from the unit Chief Examiner before doing so.

You are **prohibited from**:

- Using generative AI to generate your entire (or the majority of your) work. The work submitted must be mostly original / written by the student.
- Working with students not from your team
- Working with friends
- Working with family members

You are reminded that any work submitted must:

- represent a sincere demonstration of your human efforts, skills and subject knowledge that you will be accountable for.
- adhere to the guidelines for AI use set for the assessment task.
- reflect the University's commitment to academic integrity and ethical behaviour.

4 System Overview

This project is based on the Blackwall Tunnel (Southern Approach) which exists in the UK. We have simplified the actual system in reality to comprise of the A102, the freeway leading up to the tunnel entrance, and Tunnel Ave, the arterial road running parallel to A102 near the tunnel entrance. The main emphasis of the system is on traffic control and to prevent or minimize overheight vehicle collisions at the tunnel entrance, which only can accommodate up to 4.0m high vehicles. The system provides a overheight detection system on approach, a final overheight vehicle exit and a final overheight vehicle detection system just prior to the tunnel entrance. There are also systems in place to take control of the traffic should power fail in the area, which is run off backup batteries. You can find the feature and subsystem specifications under System Features which describes the system behaviour modelled below. US1/US2 should be capable of reading the actual vehicle height, whereas US3/US4/US4 are only required to detect an over-height vehicle passing.

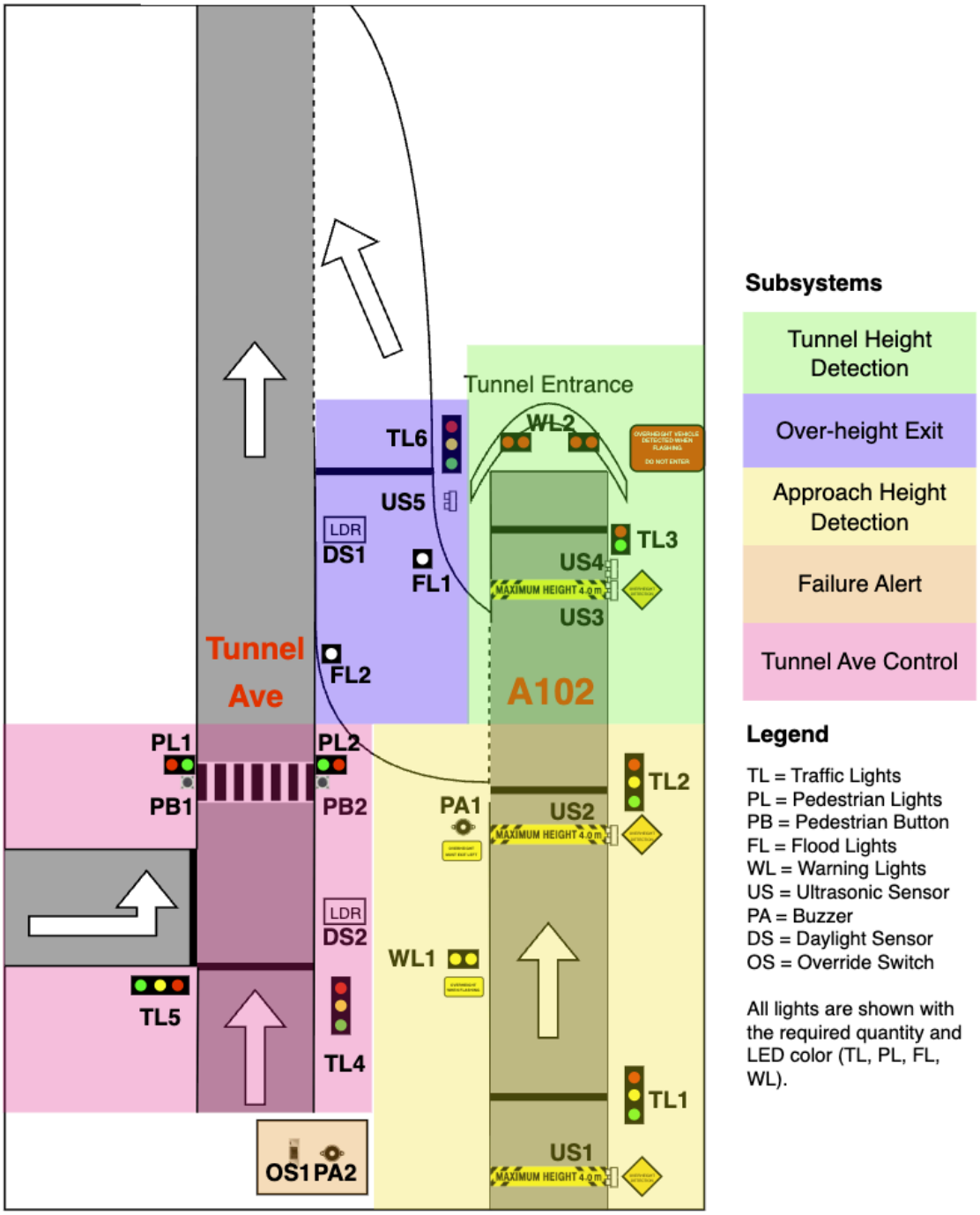

Figure 1: Model of Traffic Control Safety System for the tunnel.

5 Deliverables

The project is split into three milestones, where the first milestone is a design/plan stage; the second milestone consists of your system submission; and the third milestone is a reflection stage.

You will earn **points** over the course of this project, which will translate into marks after the completion of the full project (and completion of the viva interview). These marks will be individually scaled based on a Weighted Scaling Factor using your third ITP Metrics Peer Assessment Score (PAF3) at 50% weighting, and your Viva Interview Score at 50% weighting.

You should read section 5 carefully as it outlines your deliverables for each milestone, including all deductions and submission requirements.

The three milestones are as follows:

1. Delivery of a design document (80 points) which includes
    a. Hardware system interaction block diagram showing all interaction between hardware components as blocks
    b. High level software logical flowcharts showing all interaction between components, one flowchart for each subsystem
    c. Planned timeline of work (gantt chart)
    d. Communications plan
    e. Conflict resolution plan
2. Full system submission, demonstration and viva (oral interview) (180 points)
    a. Circuit diagrams (one circuit per function / per page as a .pdf file) for all components being demonstrated
    b. Software code (.py python files) for all work being marked
    c. Physical circuit (on demonstration day in week 11)
3. Project reflection
    a. Complete the reflection task set
    b. Complete the in-person viva in week 12

Your system **must be demonstrated in person** during the assigned session time **to achieve the maximum points available** (100%). However, suppose some part (or whole) of your prototype is not functional on the day. In this case, you use a **video your team submitted as part of your milestone on Moodle** to be marked instead for **66%** of the points available. If you have not submitted a video, your team may **explain** how your system should work for **33%** of the available points.

Individual bonus points

Knowledge gained from completing the Practical Learning Competency tasks will help you greatly throughout the project. If you complete these **within the first two weeks of the assessment offering**, you will be awarded bonus marks for each, to be applied individually to your final project mark (capped at 26%).

5.1 Milestone 1

This milestone is due at the end of Week 5 on Friday at 11:55pm.

5.1.1 Milestone Deliverables

Your team must submit a System Design Document for the **full system**.

This document includes the following sections (in this order). You are recommended to use the provided template for your submission that explains the following items. Further examples for the document are available on Moodle under Project Resources..

- System Interaction Block Diagram showing all hardware interactions for the system.
- High level flowcharts for all software features (with hardware interactions) for the system.
    - These flowcharts should include a short description that lists all the features being implemented.
- Proposed Timeline for work showing scheduled time to work on the whole project across the semester. Your timeline should include a formal check in with your demonstrator during one of the practical classes for informal feedback on your team implementation of required features.
- Communications plan showing your team's primary, secondary and backup communications methods, along with the expected response time in varying scenarios. You should also decide on a template to use for meeting minutes - which is assessed in M3.
- Conflict resolution plan showing your team's resolution and escalation process for disagreements and conflicts, including when work is not done to the agreed standard, or if work is not done on time.

5.1.2 Milestone Weighting

This milestone is marked out of 80 points. Points value breakdown as follows:

- System Interaction Block Diagram: 20 points (one page)
  Marking criteria:
  - Includes all hardware components for features that the team will build
  - Components clearly belong to a specific subsystem on the diagram
  - Hardware components are logically linked to each other appropriately
- High Level Flowcharts: 30 points
  Marking criteria:
  - Items in flowchart are at a high level (one flowchart per subsystem)
  - Proposed flow of logic is sensible
  - Includes all relevant software components for features that the team will build
- Proposed Timeline: 10 points
  Marking criteria:
  - Timeline shows proposed progression of work for all features that the team will build
  - Timeline shows work breakdown by member (assigns work to each member fairly)
  - Timeline includes check in with demonstrator for informal feedback on required features
- Communications Plan: 10 points (expected 1 - 2 pages)
  Marking criteria:
  - Plan shows primary, secondary and backup comms methods
  - Plan includes expected response times in more than one sensible scenario
- Conflict Resolution Plan: 10 points (expected 1.25 - 2 pages)
  Marking criteria:
  - Plan shows team resolution/escalation plan for resolving conflicts and disagreements
  - Plan shows team plan for handling low quality or incomplete / late work

Please note marks for the milestone will not be finalised until completion of the full project, after application of the Viva and PAF scaling (see section 4.2.6.3 and 4.3 for more details on scaling).

Deductions apply. Refer to section 5.1.4

5.1.3 Submission Format

You will need to submit two pdf files. Please **pay careful attention to the filenames required**.

1. **AI Usage Declaration**: a PDF declaring your team's use of generative AI on the project. At minimum, this should contain declarations of what your team has used genAI for. If you have not used genAI, state that your team has not used genAI at all.
   Name this file TeamNNN-M1_AIDeclaration.pdf, replacing NNN with your team code, i.e. if your team is MS02, your filename should be **TeamMS02-M1_AIDeclaration.pdf**
2. **Design Document**: Name this file TeamNNN-M1_SDD.pdf, replacing NNN with your team code, i.e. if your team is MS02, your filename should be **TeamMS02-M1_SDD.pdf**
3. If you submit the wrong files,
   - If it's before the deadline, post on the ed forums and we will ask you to resubmit.
   - If it's after the deadline, email eng1013.clayton-x@monash.edu the files and we will ask you to resubmit on Monday.
4. A penalty applies if the filenames are incorrect. We can refuse to mark the files if provided in an incorrect format.

5.1.4 Deductions

Category	Description	Deduction
Overall Submission	Incorrect submission format (filetype not PDF, structure) or left instructions in template (didn't delete blue text) Incorrect **filename format**.	-3
System interaction block diagram quality	System Interaction Block diagram for hardware is of poor quality	-3
High level flowchart quality	High level flowchart for software is of poor quality, lacking clarity, completeness, or accuracy in its representation. (includes readability / diagram flowing across pages in a jigsaw puzzle, wrong flowchart shapes, or have parallel logic)	-3
Proposed Timeline quality	Proposed Timeline incomplete or poor quality, lacking thoroughness, clear milestones, or a realistic allocation of tasks and deadlines. Timeline is missing scheduled check-in for feedback.	-3
Comms Plan quality	Plan is incomplete or of poor quality (brief, isn't detailed) Plan is missing a meeting minutes template.	-2
Conflict Plan quality	Plan is incomplete or of poor quality (brief, isn't detailed)	-2

5.2 Milestone 2

Milestone 2 is due at the end of Week 10 on Friday at 11:55pm, with demonstration in Week 11, and viva (Milestone 3) in Week 12.

5.2.1 Milestone Deliverables

Your team must demonstrate a functioning system (180 points) that consists of the following:

- Code and circuits implementing required features (in blue) of following subsystems (80 points):
  - Subsystem 1 (Approach Height Detection Subsystem)
  - Subsystem 2 (Tunnel Ave Control Subsystem)
  - Subsystem 3 (Over-height Exit Subsystem)
  - Subsystem 4 (Tunnel Height Detection Subsystem)
  - Subsystem 5 (Failure Alert Subsystem) (*note, this is a hardware only subsystem*)
- Additional code and circuits implementing general and/or integration features from the five subsystems as selected (80 points). A point cap applies to each subsystem as follows for general and integration features selected for implementation (caps apply based on where the feature is specified):
  - Subsystem 1 (Approach Height Detection Subsystem): 10 points
  - Subsystem 2 (Tunnel Ave Control Subsystem): 20 points
  - Subsystem 3 (Over-height Exit Subsystem): 10 points
  - Subsystem 4 (Tunnel Height Detection Subsystem): 20 points
  - Subsystem 5 (Failure Alert Subsystem): 20 points
- Circuit Diagrams for all implemented features. You will not be allowed to demonstrate circuits for which there are no circuit diagrams submitted.
- Full system integration (of all subsystems) as a single code-base (meaning a single code launch execution point) and operating on a single Arduino (20 points).

Points caps are applied to each subsystem's general and integration features - you can implement more than the cap amount of points but you will at most earn that cap amount from the subsystem.

Deductions will only be applied after the total points earned is calculated.

Note that for full system integration marks (20 points), your code must execute from one single starting point (i.e. running the main.py file will start the entire system) and run off a single Arduino for your project demonstration. Any subsystems that are partially integrated will be assessed as a single unit (marked together).

Your team must demonstrate the built system's functionality to your demonstrator in your Week 11 class.

5.2.2 Milestone Weighting

This milestone is marked out of 180 points.

Marking Rubric for features:

- Functional and working without errors: 100% of points awarded
- Contains errors but can demonstrate was working in submitted video: 66% of points awarded
- Contains errors but can explain how to fix: 33% of points awarded
- Required feature not attempted: 33% of points deducted
- Attempted but does not work, cannot explain: 0% of points awarded

Please note marks for the milestone will not be finalised until completion of the full project, after application of the Viva and PAF scaling (see section 4.2.6.3 and 4.3 for more details on scaling).

Deductions apply. Refer to section 5.2.5.

5.2.3 Submission Format

You will need to submit everything as a single **zip** file. Please pay careful attention to the filenames required.

Your zip file should contain the following files:

```
TeamNNN/
        TeamNNN-M2_AIDeclaration.pdf
        circuit_diagrams.pdf
        (multiple files of any sensible name).py ← ensure your filenames contains no spaces
        backup_video_url.txt
```

- **AI Usage Declaration**: a PDF declaring your team's use of generative AI on the project. At minimum, this should contain declarations of what your team has used genAI for. If you have not used genAI, state that your team has not used genAI at all.
  Name this file TeamNNN-M2_AIDeclaration.pdf, replacing NNN with your team code, i.e. if your team is MS02, your filename should be **TeamMS02-M2_AIDeclaration.pdf**
- **Circuit Diagrams:** The circuit diagrams must be in PDF format (.pdf), in a single file containing one circuit per page, and can be landscape A4 or portrait A4. You should have one circuit for each logical component.
- **Physical Circuit**: You will need to bring the physical circuit on the day of your practical in Week 11 for demonstration and marking of M2. Please note that you are required to cut, and strip all wires (solid core) for your circuit. You can keep jumper wires (provided in your kit) for connections from the circuit to the Arduino. If the wires are too short, it is recommended that you cut longer ones from the lab-provided wire. You will have access to this wire and wire stripping in week 10's class.
- **Code:** The code submitted must be in python format (.py), and can be split across multiple files if needed. Please ensure that the filenames contain no spaces.
- **Backup Videos:** Teams may choose to include a set of short video clips in .mp4 format demonstrating each prototype functionality being assessed. You should upload the video to a folder in your assigned team google drive folder, and include the URLs of each video in a text file in your moodle submission zip. This will serve as a backup for assessment if the demonstration during class fails. The filename should be titled backup_video_url.txt
- **Submission naming format:** Put your project files following the structure above in a folder named TeamNNN, replacing NNN with the code for your team. Then ZIP the folder up and name it TeamNNN-M2.zip, replacing NNN with the code for your team, i.e. if your team is MS02, your filename should be TeamMS02-M2.zip (max zip file size 10MB).
- Please note that we will not mark files that are not in the correct forms specified.
- A penalty applies if the filenames are incorrect.
- If you submit the wrong files,
  - If it's before the deadline, post on the ed forums and we will ask you to resubmit.
  - If it's after the deadline, email eng1013.clayton-x@monash.edu the files and we will ask you to resubmit on Monday.

5.2.4 Testing

All teams should ensure that they arrive on-time for their practical class session.

You will have 30-35 minutes for demonstration, so your group will need to arrive a minimum of ten minutes before your presentation time to prepare and finalise any set up. You should come to class with **your project fully assembled** as no time is given to build your circuits.

5.2.5 Deductions

The client defines an “inappropriate current” as a current that is either too low for consistent operation or too high for the components used (risking damage).

Category	Description	Deduction
Submission	Incorrect filename submitted, or incorrect folder structure.	-5
Submission	Incorrect format submitted	Resubmission required before marking will commence.
Overall	No circuit diagram submitted for a feature.	Affected feature will not be assessed (no attempt given)
Overall	Other hardware components besides the allowed hardware components are used (items not provided in your project kit).	The project will be marked once the offending hardware components are removed - they are not to be replaced.
Overall	Non permitted packages, methods and techniques used.	Offending code to be commented out.
Circuits	Circuit Diagram does not match physical circuit submitted	Cap of feature mark to 33%
Probe Points and Appropriate Current	● At least one current probe point is included for at least one LED and at least one buzzer (pick loudest / brightest) ● Inappropriate current for any LEDs or buzzers	-5
General Code Quality	● Console print statements are unclear, have insufficient information, or are poorly formatted. ● Console alerts are at an inappropriate rate (i.e. spams the console / prints multiple times per second). If code is used to clear the console, this deduction will automatically apply. ● Code crashes, does not exit cleanly, has an unhandled exception, uses unspecific exception handling, or exits with an error message.	-5
Physical Circuit Quality	● Physical circuit doesn’t follow wire colour conventions, ● Physical circuit doesn’t use power/ground rails appropriately (components directly connected to a rail) ● Physical circuit uses spaghetti wiring (except for wires to the Arduino).	-5
Circuit Diagram Quality	● Circuit Diagram doesn’t meet required standards ● Circuit Diagram components are incorrectly oriented. ● Circuit Diagram is not correctly labelled with unique identifiers and values. ● Circuit Diagram uses wrong component shapes (must use shapes from the library provided).	-5
Coding Quality	● Code doesn’t meet coding standards ● TODOs left in code ● Variables not in lowerCamelCase ● Functions not in snake_case ● Indentation not 4sp = 1 tab ● Function header documentation in wrong location, format, incomplete, missing or not present ● File header documentation in wrong format, incomplete, missing or not present	-5

5.3 Milestone 3

Milestone 3 is due at the end of Week 11 on Friday at 11:55pm. This submission will unlock at the beginning of Week 11.

5.3.1 Milestone Deliverables

Each student must submit a personal reflection to Moodle by completing the Project Reflection Quiz. As part of this assessment, you will be checked on your project management. You will be required to submit your team meeting minutes to the quiz. Please note that you will need to individually submit the meeting minutes to the quiz. You are not allowed to use generative AI on this milestone task.

5.3.2 Milestone Weighting

This milestone is **unweighted**, but is a requirement prior to completing the project viva. Students who do not complete the reflection quiz will not be able to complete their viva, and thus receive 0 marks for the project. You will be asked to complete the reflection on the spot in order to do your viva.

Deductions apply. Refer to section 5.3.4

5.3.3 Viva (Interview)

The viva is an individual interview in week 12 to confirm each team member's understanding of the built system. Each member of the team will be individually interviewed about what has been developed. You must have completed the reflection quiz prior to being allowed to take the viva. You will be required to answer questions based on both the software and hardware components the group has developed. Thus it is important to make sure you understand all parts of the prototype, even sections you did not individually work on.

Total project marks will be scaled based on the level of understanding demonstrated in the Viva.

If you miss (do not attend) your project interview, you will not receive any project marks.

Category Description	Detailed Description
Complete Understanding	The student has **clearly prepared** and **understands the code/circuits**. They can answer the questions correctly and concisely with little to no prompting.
Tolerable understanding	The student is **reasonably well prepared** and can **consistently provide answers** that are **mostly correct**, possibly with some prompting. The student may lack confidence or speed in answering.
Selective understanding	The student gives answers that are **partially correct or can answer questions about one area correctly** but another not at all. The student has not prepared sufficiently.
Trivial understanding	The student may have seen the code/circuits before, and can answer **something partially relevant or correct** to a question but they **clearly can't engage in a serious discussion of the work**.
No Understanding	The student has **not prepared**, **cannot answer even the most basic questions** and likely has not even seen the code/circuits before, or did not attend the interview.
Absent	The student has **not attended the interview** for the project.

5.3.4 Deductions

Category	Description	Deduction
Submission	No submission is made	Viva will not be conducted, and a zero will be awarded unless the student completes the submission.
Project Management	No meeting minutes submitted.	-10
Project Management	Meeting minutes are submitted, but of poor quality, or are lacking in detail	-5
Reflection	Student has not provided a genuine reflection.	-5

5.4 ITP Metrics Peer Assessment

We monitor student team health via ITPMetrics.

There are three ITPMetrics surveys, each closing after 5 working days.

PAF1: Opens Week 4 Fri, Closing Week 5 Fri.

PAF2: Opens Week 7 Fri, Closing Week 8 Fri.

PAF3: Opens Week 12 Fri, Closing Week 13 Fri.

During the PAF weeks, students will be emailed, and required to login to https://monash.itpmetrics.com/ and complete a peer-assessment of their teammates, to give feedback to each other on their teamwork performance.

This comes in the form of rating your peers on some aspects of teamwork, and providing some comments. Once this is done, the system will compile and generate some written feedback to students, which you should discuss in the next team meeting and document in your team meeting minutes. These should be used to improve your behaviour and performance going forward. You will explore these aspects of teamwork much more deeply in ENG1012.

The first two peer assessment factors (PAF1/PAF2) are used solely for monitoring team health.

We will use the final peer assessment factor (PAF3) as a scaling factor in your project. PAF3 will be used to scale the total marks of the project to indicate overall contribution.

The nominal scale for PAF can range from 0 to 1.1 and is based on your teammate's ratings of your performance relative to the average rating of the entire team. Essentially, if everyone is happy, everyone receives a PAF of 1. If a team member receives a consistently higher or lower rating than the average of the team, they would receive a PAF of higher or lower than 1.

Teams that are under or have only 3 members and score a PAF of at least 1.0 will be scaled up to PAF of 1.2.

Students that do not complete their ITP Metrics peer assessments will receive a maximum PAF of 0.9.

Please ensure that if you encounter problems, to reach out to the unit lecturers as soon as possible for early intervention.

6 Milestone Implementation TL;DR

This project welcomes students of all skill levels, from beginners to experienced programmers and electronics enthusiasts. ***This project can be completed with the knowledge attained by attempting all the lab exercises covered in the Practical classes***.

Achieving a Credit or Distinction is straightforward by consistently completing weekly lab exercises and adapting them to your project.

Achieving Higher Marks (>80%)

To score above 80%, you'll need to tackle the **system integration features, or attempt complete system integration**. These are designed to be very challenging, particularly for coding and electronics enthusiasts, and will require a significant time commitment (approximately 20-30 hours for beginners).

Time Management and Bonus Marks

Students who opt out of integration will save a considerable amount of time – a valuable resource at university. Instead, you might consider pursuing the **bonus marks for Practical Learning Competencies** (up to 2% = 20 points each). These can be earned within the first two weeks of the assessment offering and will individually contribute to your final project mark (capped at 26%), potentially compensating for not attempting integration.

Integration: A Challenging Endeavor

Warning: Complete Integration is difficult and can be extremely time-consuming!

We recommend teams only work on integration if they are competent in their abilities. If complete integration isn't pursued, students must be able to demonstrate the features of each subsystem independently.

Levels of Integration

There are two levels of integration for this project:

- **Partial Integration**: Points are awarded for integrating specific subsystems and implementing their associated features. For these features, the integrated subsystems must run off a **single Arduino**, both in software and hardware.
- **Complete System Integration**: This requires all subsystems to interact with each other, regardless of the specific integration features your team chooses to implement. This means your team must use a **single Arduino and codebase** for the entire project.

Using a single Arduino necessitates careful decisions regarding pin selection and usage. As demonstrated in practice classes and workshops, you can use multiplexers like shift registers (for outputs) or an R-2R ladder (for inputs) to manage pin usage effectively.

Important Considerations for Integration

It's common for existing functionality to "break" when integrating multiple subsystems, often due to rushing the process. This can negatively impact your marks. Therefore, it's best to **avoid attempting integration in the final days before submission**, as last-minute efforts often lead to accidental system malfunctions.

Core Demonstration Expectation

Regardless of the integration level, the fundamental expectation is that your team can demonstrate each subsystem on its own (assuming no integration was performed). If your team has implemented integration features, those subsystems must work together (launch from the same code, use the same Arduino) as described in the feature.

Seek Guidance

If you have any doubts about your team's implementation plan, discuss it with your team manager well in advance.

7 System Features

This section outlines the functional requirements of the system, divided into five subsystems.

The system should continue to run, uninterrupted, until the user chooses to exit (via KeyboardInterrupt).
You must handle the KeyboardInterrupt and cleanly turn off all pins, then shutdown the Arduino.

For your convenience, the features in each subsystem are color coded as follows:

- Required features are shown in **blue highlights**.
- General features are shown in normal text (no highlight).
- Integration features are shown in **yellow highlights**.

7.1 Approach Height Detection Subsystem

This is the primary overheight detection system, consisting of two sets of overheight detection ultrasonic sensor (US1, US2), two sets of red/yellow/green traffic lights (TL1, TL2) spaced 500m apart along the road, a speaker system (PA1) and a set of two yellow warning lights (WL1). The subsystem is the yellow highlighted section of the traffic control system prior to the over-height exit on the system overview.

Code	Description	Value
1.R1	Upon detection of an overheight vehicle by US1, an alert is printed to the console. The alert should include the detected height of the vehicle, as well as the date/time of detection.	5
1.R2	Upon detection of an overheight vehicle by US1, the following sequence is triggered: 1. TL1 switches to yellow for 1s, then to red for 30s. 2. TL1 then turns back to green.	5
1.R3	Upon detection of an overheight vehicle by US2, the following sequence is triggered: ● If US1 did <u>not</u> detect an overheight vehicle, TL1 and TL2 immediately switch to yellow for 1s, then red for 30s before turning back to green. ● If US1 <u>detected</u> an overheight vehicle, then TL2 switches to yellow for 1s, then red for 30s before turning back to green. You will need to calculate a reasonable threshold for determining if it's the same vehicle based on the sensors being 500m apart for a heavy vehicle going at highway speeds.	5
1.R4	Upon system start-up, the user is prompted to enter the overheight limit configuration for the system. If the user hits enter without providing any input, it defaults to the system default value of 4.0 meters. This value should be used as a variable in the entire subsystem.	5
1.G1	Upon detection of an overheight vehicle by US1/US2, WL1 turns on and flashes each yellow LED in turn at 2-5 Hz so long as either TL1 or TL2 are not green.	5
1.G2	Upon detection of an overheight vehicle by US1/US2, PA1 turns on and begins sounding a unique buzzer tone (400Hz - 800Hz). The buzzer tone must be generated using a 555 timer. PA1 turns off if TL1 and TL2 are no longer red.	10
1.G3	If US2 continues to detect an overheight vehicle, hold TL1/TL2 at red light. After 30s of TL2 at red, the buzzer tone on PA1 changes to a higher frequency tone (2kHz - 4kHz). The buzzer tone must be generated using a 555 timer.	10
1.G4	The data feed from ultrasonic sensor US1/US2 is filtered for noise appropriately using a moving average.	5
1.I1	**This feature requires integration with <u>subsystem three</u>.** *This feature overrides the default behaviour above.* When TL1 and TL2 turn red as controlled by US1/US2, they will <u>stay red</u> until US5 detects an overheight vehicle passing (hence exiting the system), and US1/US2 and US5 no longer detect an overheight vehicle. Then, set TL1 and TL2 back to green, and turn off WL1 and PA1.	10

7.2 Tunnel Ave Control Subsystem

This subsystem contains a major road, the Tunnel Ave with a minor controlled junction that turns into it via controlled Traffic Lights (TL4/TL5). It has a pedestrian crossing with its associated red/green pedestrian lights (PL1/PL2) and pair of pedestrian crossing push buttons (PB1/PB2) on the Tunnel Ave arterial road. DS2 is a LDR that controls the traffic light timing for day/night cycles. This subsystem is highlighted in pink on the system overview.

Category	Description	Value
2.R1	If the pedestrian push button PB1/PB2 is pressed, the following sequence should be triggered after a two second wait. If TL5 is currently not red, TL5 goes yellow for 3s, then red. Otherwise, TL4 turns yellow for 3 seconds, then red. 1. Then, PL1/PL2 turns green for 3s, then flashing red for 2s before resetting to solid red. 2. Then, TL4 turns back to green.	10
2.R2	If the pedestrian push button PB1/PB2 is pressed, show this information on the console once (until the sequence 2.R1 is run). (i.e. you should ensure that holding down or spamming the push button does not trigger multiple console prints).	5
2.R3	TL4 and TL5 operate on a 20-10 second cycle. That is, after 20 seconds on green, TL4 will turn yellow (3s), then red. When TL4 turns red, TL5 turns green for 10 seconds, then yellow (3s), then red, and the cycle repeats with TL4 turning green.	5
2.G1	After PB1/PB2 is pressed and 2.R1 runs, enforce a minimum wait of 30s before running 2.R1 again. The pedestrian push button PB1/PB2 is explicitly debounced using hardware components such that a clear signal is sent.	5
2.G2	The frequency of flashing of PL1/PL2 red light is generated using a single 555 timer.	5
2.G3	LDR DS2 is used to detect if it's currently day or night. If it's night-time, then the traffic light timings for TL4 and TL5 are overridden as follows: • TL4 green cycle is now 30s. • TL5 green cycle is now 5s. • If PB1/PB2 is triggered during any cycle, both TL4 and TL5 will follow their respective timings and stop at red for 5s to allow PL1/PL2 to follow their pedestrian cycle of green/flashing red.	5
2.I1	**This feature requires integration with <u>subsystem three</u>.** *This feature overrides the default behaviour.* Upon detection of a vehicle by US5, if TL4 is currently not red, it should turn yellow (3s) then red. If TL5 is currently not red, it should turn yellow, then red. Then, PL1/PL2 turns green. When a vehicle is no longer detected by US5, the following sequence should execute: • PL1/PL2 should turn flashing red for 2s before resetting to solid red. • Then, TL4 should turn green. • TL4/TL5 should then resume the normal 2.R3 cycle.	10

7.3 Over-height Exit Subsystem

This is the subsystem controlling the set of the red/yellow/green traffic lights (TL6), a light dependent resistor for detecting day/night (DS1), a set of two floodlights (white LEDs) (FL1, FL2) as well as the vehicle presence detection ultrasonic sensor (US5) on the over-height exit off the A102. This subsystem is highlighted in purple on the system overview.

Category	Description	Value
3.R1	Upon detection of an overheight vehicle by US5, the following sequence should trigger to allow the vehicle to exit: 1. TL6 turns green for 5s 2. TL6 then turns yellow for 3s 3. TL6 then turns back to red	10
3.R2	If the ultrasonic sensor US5 continues to detect an overheight vehicle after TL6 has been green for 5s, TL6 continues to stay green until US5 no longer detects a vehicle.	5
3.G1	If the ultrasonic sensor US5 continues to detect an overheight vehicle after TL6 has stayed green for at least 5s, the following sequence runs instead: 1. TL6 flashes green at 2-5 Hz continuously until US5 no longer detects an overheight vehicle 2. TL6 turns back to red (otherwise (1) will continue). The frequency of flashing green of TL6 at 2-5Hz is generated using a 555 timer.	10
3.G2	The data feed from ultrasonic sensor US5 is filtered for noise appropriately using a moving average.	3
3.G3	If the ultrasonic sensor US5 detects an overheight vehicle and the LDR DS1 detects that it is night time, the following events occur: 1. Turn FL1 and FL2 on and this stays on while US5 detects an overheight vehicle 2. Turn FL1 and FL2 off afterwards/after the condition is cleared	5
3.G4	If the LDR DS1 detects that it is night time, the base green timing for TL6 is changed from 5s to 10s. This affects all triggers relying on TL6 green such as in 3.R2 and 3.G1.	5
3.I1	**This feature requires integration with subsystem two.** *This feature overrides the default behaviour.* Upon detection of an overheight vehicle by US5, 3.R1 sequence is delayed by a minimum of 3s (upper limit delay for TL6 is determined by the time needed for TL4/TL5 to complete sequences and turn red) to allow for the Tunnel Ave traffic lights to turn red before allowing the exit traffic to move.	5

7.4 Tunnel Height Detection Subsystem

This is the “last chance” overheight detection system, consisting of a set of overheight detection ultrasonic sensor (US3/US4), a set of red/green traffic lights (TL3), and two sets of two red warning lights (WL2) which are used to indicate tunnel closure when flashing. This subsystem is highlighted in green on the system overview.

Category	Description	Value
4.R1	Upon detection of an overheight vehicle by US3, TL3 should turn from green to red immediately. The system can only reset to ‘normal state’ if an overheight vehicle is no longer detected by US3. (i.e. TL3 turns back to green)	5
4.R2	The ultrasonic sensor US4 is used to verify data read by US3. US3 is only triggered if US4 reads the same value (within an acceptable error range).	5
4.R3	Upon system start-up, the user is prompted to enter the overheight limit configuration for the system. If the user hits enter without providing any input, it defaults to the system default value of 4.0 meters. This value should be used as a variable in the entire subsystem.	5
4.G1	Upon detection of an overheight vehicle by US3, WL2 turns on and flashes each red LED in turn at 2-5 Hz so long as US3/US4 detects an overheight vehicle. The pattern of flashing is XOXO then OXOX where WL2 is configured as two pairs of red LEDs.	5
4.G2	(Must implement 4.G1) WL2 flashing red frequency is generated electrically using a 555 timer.	5
4.I1	**This feature requires integration with <u>subsystem one</u>.** *This feature overrides the default behaviour from subsystem 1.* Upon detection of an overheight vehicle by US3/US4, execute the following sequences until US3/US4 no longer detects an overheight vehicle: ● TL1 and TL2 turns red ● PA1 starts sounding a unique tone between 4kHz and 6kHz. This frequency is generated using a 555 timer. ● WL1 turns on and flashes each yellow LED in turn at 5-10 Hz	10
4.I2	**This feature requires integration with <u>subsystem two</u>.** *This feature overrides the default behaviour from subsystem 2.* Upon detection of an overheight vehicle by US3/US4, execute the following sequences until US3/US4 no longer detects an overheight vehicle: ● TL4/TL5 turns red (overriding any existing state temporarily).	5

4.I3	**This feature requires integration with <u>subsystem one, subsystem two and subsystem three</u>.**
	
	This feature overrides the default behaviour from subsystems 1, 2 and 3.
	Upon detection of an overheight vehicle by US3/US4, execute the following alternate behaviour:
	
	● The system reset to ‘normal state’ now relies on:
	  ○ US5 detecting an overheight vehicle exiting
	  ○ US1/US2/US3/US4/US5 no longer detecting an overheight vehicle
		10

7.5 Failure Alert Subsystem (Advanced)

This is the subsystem that operates in case of power failure to the Tunnel Height Detection Subsystem in order to shut down the tunnel for safety reasons. It consists primarily of a backup battery pack, an override switch (OS1), a speaker system (PA2), and has overarching control of all the red traffic lights on the A102. This system must be built entirely in hardware and is electrically controlled (i.e. no software). This subsystem is highlighted in orange on the system overview. Note for testing: We will not test 'resumption' of existing states on power restoration.

The **general features** (anything above 5.R1) of this subsystem are significantly more difficult to implement than all the others, and **should only be attempted by students that have mastered the electrical content**. You will need to have completed the last week of lab activities. This subsystem is only worth 10 + 10 = 20 points, so you can attain these marks alternatively by completing the Practical Learning Competencies during the bonus mark periods.

Category	Description	Value
5.R1	This subsystem is powered by an external battery supply and must not be powered by the Arduino's pins. Upon detecting a loss of power to the Tunnel Height Detection Subsystem's Arduino (by monitoring the Arduino's power pins), PA2 should start sounding an alert tone between 500Hz and 2kHz immediately. A 555 timer must be used to generate the tone. The alert tone should be clearly audible (not soft) but not excessive (not too loud).	10
5.G1	The subsystem maintains the alert tone for 3s after power is restored, after a power outage has occurred. (This means the tone should continue to sound for 3s after power is restored after a power outage, and not impact the immediate start of the alert tone upon detection of a power outage).	5
5.G2	If the override switch OS1 has been turned on, enable the Failure Alert subsystem (act as if a power outage has been detected). The override switch OS1 is debounced using hardware components such that a clear signal is sent.	2
5.G3	The detection circuit (monitoring power loss from the Arduino) is done by implementing a comparator op-amp.	5
5.I1	**This feature requires integration with <u>subsystem one</u>.** *This feature overrides the default behaviour.* If the Failure Alert subsystem is triggered (on a power outage), run: • TL1 and TL2 turns red until power is restored (this interrupts any existing sequence). • On power restoration, resume the original sequence.	8
5.I2	**This feature requires integration with <u>subsystem four</u>.** *This feature overrides the default behaviour.* If the Failure Alert subsystem is triggered (on a power outage), run: • TL3 turns red until power is restored • (Must implement 4.G1) WL2 turns on and flashes each red LED in turn at 5-10 Hz using a 555 timer to generate the flashing signal until power is restored. • On power restoration, resume the original sequence.	8